**MA3521 — Separation of Variables**

Name:

**Problem 1** $y' = e^{t+y^2}/y, \qquad y(1) = 2$

**1a. Identify the factors in $y' = g(t)h(y)$.**

$g(t) =$ $\qquad$ $h(y) =$

**1b. Write the integrals. Do not evaluate.**

$\int$ $\, dy = \int$ $\, dt + C$

**1c. Evaluate the constant.**

$C =$

**Problem 2** $y' = e^{t+y}, \qquad y(0) = 0$

**2a. Identify the factors in $y' = g(t)h(y)$.**

$g(t) =$ $\boxed{\phantom{xxxxxxxxxx}}$ $\qquad h(y) =$ $\boxed{\phantom{xxxxxxxxxx}}$

**2b. Write the integrals. Do not evaluate.**

$\int \boxed{\phantom{xxxxxxxxxx}}\, dy = \int \boxed{\phantom{xxxxxxxxxx}}\, dt + C$

**2c. Evaluate the constant.**

$C =$ $\boxed{\phantom{xxxxxxxxxx}}$